Updated on 25/05/2026

BITP 3513 ADVANCED DATABASE PROGRAMMING

INDIVIDUAL ASSIGNMENT 2-a (10%)

Read the case study below and answer the following question.

Case Study: Sales Performance Monitoring System

A small retail store records product sales information in a database. The store owner wants an Oracle APEX application to analyse sales performance, automate notifications, send emails, and transfer the application data to another environment.

Data Dictionary of Product_Sales		
Attribute	**Data Type**	**Description**
SALE_ID	NUMBER	Unique sales transaction ID (Primary Key)
PRODUCT_NAME	VARCHAR2(100)	Product name
CATEGORY	VARCHAR2(50)	Product category (Electronics, Stationery, Food, etc.)
QUANTITY_SOLD	NUMBER	Quantity sold
UNIT_PRICE	NUMBER(10,2)	Price per unit
TOTAL_SALES	NUMBER(10,2)	Total sales amount
SALE_DATE	DATE	Date of sale
BRANCH_NAME	VARCHAR2(50)	Branch where the sale occurred
STATUS	VARCHAR2(20)	Status updated by Automation (e.g., Normal, Top Seller)

Product_Sales.sql

```
CREATE TABLE PRODUCT_SALES (
    SALE_ID NUMBER PRIMARY KEY,
    PRODUCT_NAME VARCHAR2(100),
    CATEGORY VARCHAR2(50),
    QUANTITY_SOLD NUMBER,
    UNIT_PRICE NUMBER(10,2),
    TOTAL_SALES NUMBER(10,2),
    SALE_DATE DATE,
    BRANCH_NAME VARCHAR2(50),
    STATUS VARCHAR2(20)
);
```

Based on case study, the information of data dictionary and SQL code were provided. There are FOUR (4) tasks. Answer all questions by follow the step given for each task. Provide short explanation for each screenshot.

Prepared by Dr Nur Atikah Arbain

Task 1: Create Charts
1. Create a table named PRODUCT_SALES. 2. Insert at least 30 sales records. 3. Create an Interactive Report to display the sales data. 4. Create ONE (1) suitable charts to visualise the sales information. 5. Provide a screenshot of the charts and table of sales reports.

Task 2: Create an Automation
1. Create an Automation that runs on the PRODUCT_SALES table. 2. Configure the Automation to update a status field or generate a summary record. 3. Execute the Automation. 4. Review the Automation Execution Log. 5. Provide a screenshot of the Automation configuration and Automation Execution Log.

Task 3: Create a Send Email Process
1. Create a page that allows users to submit sales information. 2. Configure a Send Email process to send a notification after form submission. 3. Test the email process. 4. Provide a screenshot of the Form Page, Send Email process and email queue or received email.

Task 4: Migrate Database Objects and Data
1. Export the PRODUCT_SALES table and its data. 2. Import the exported file into another workspace or schema. 3. Verify that the table and records are successfully imported. 4. Provide a screenshot of the Export process, Import process and showing the imported records.

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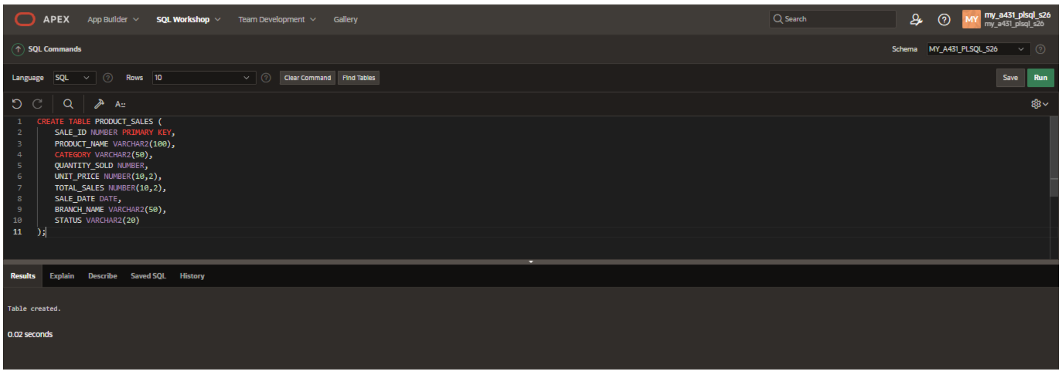
Lab Session: LAB THURSDAY 4-6

Answer:

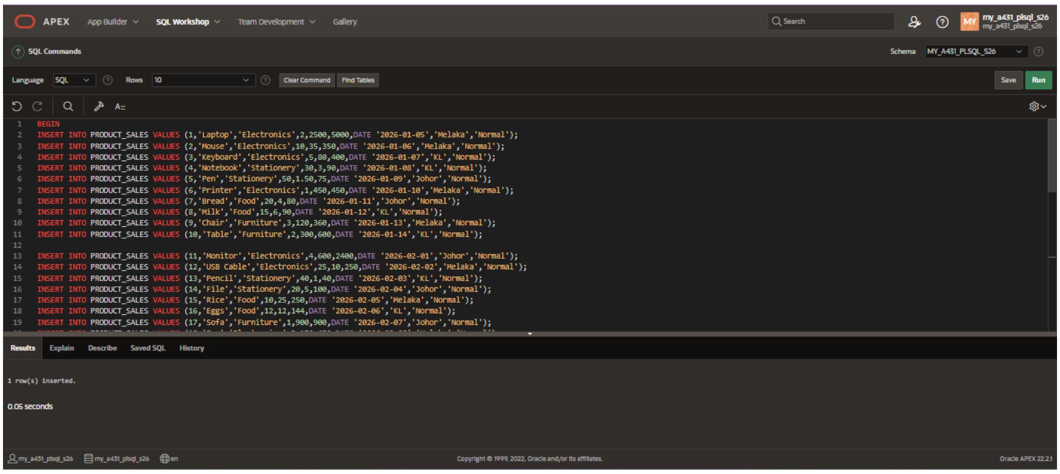
TASK 1 : Create Charts

a. **Create table** PRODUCT_SALES.



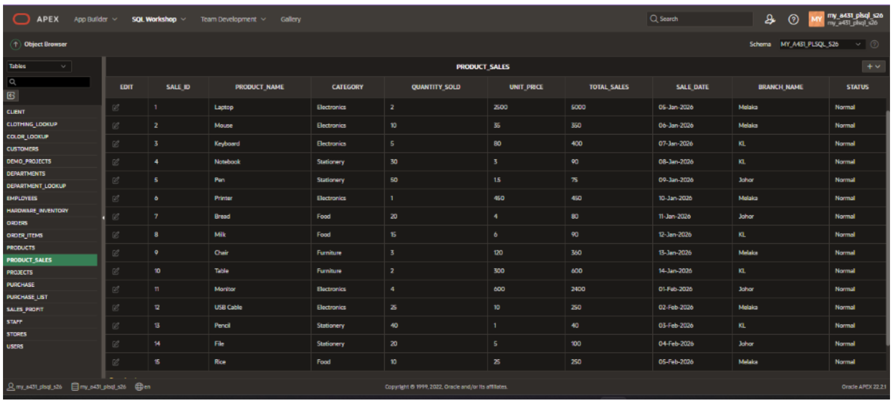
The PRODUCT_SALES table was created in Oracle Database based on the given data dictionary. The table stores sales transaction information including product details, sales amount, and status.

b. **Insert Value**

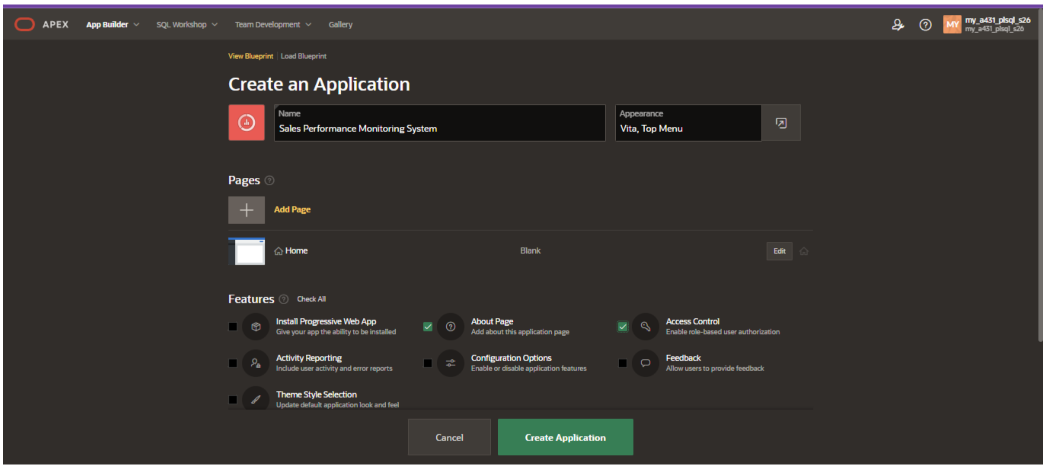


A total of more than 30 sales records were inserted into the PRODUCT_SALES table to provide sufficient data for analysis and reporting.

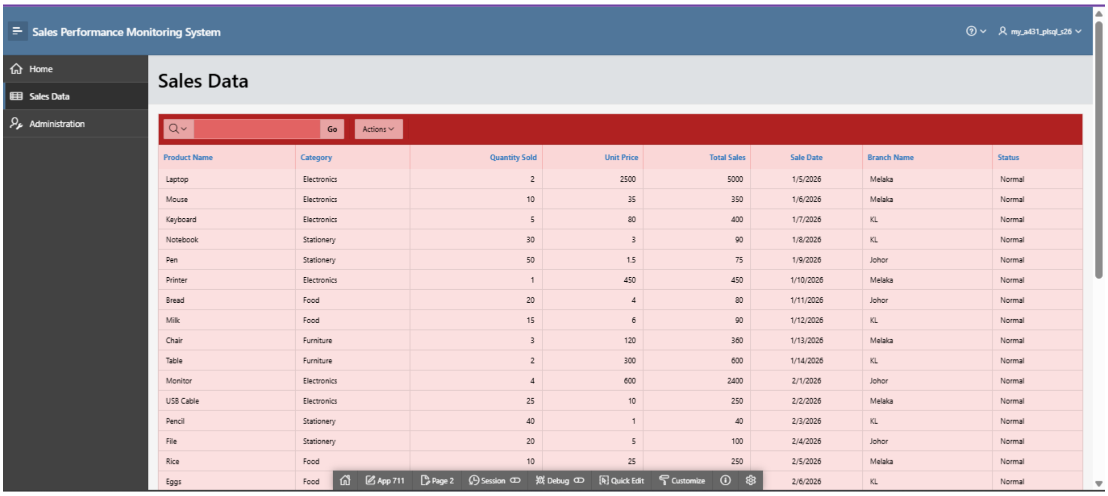
Result :



c. Create Application + Interactive Report

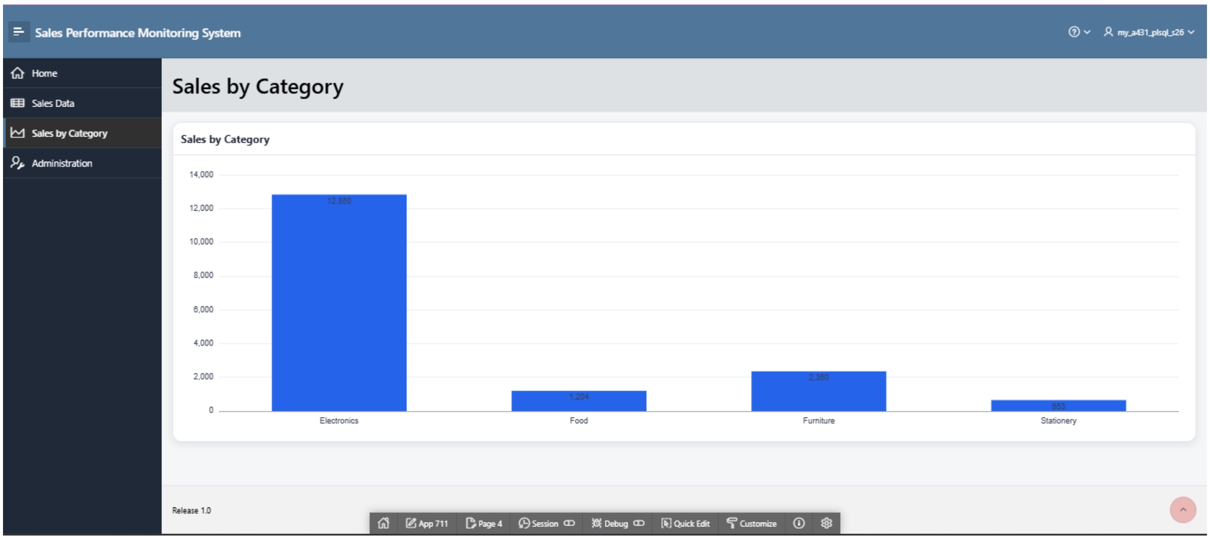


I. Interactive Report – sales product



An Interactive Report was created to display all sales records stored in the PRODUCT_SALES table. Users can search, sort, and filter the sales data easily.

ii. Chart



A bar chart was created to visualise total sales by product category. The chart helps users identify which category generates the highest sales.

Summary task 1 :

<u>In this task, the PRODUCT_SALES table was created and populated with sales records. An Interactive Report and a bar chart were developed in Oracle APEX to display and analyse sales performance by category.</u>

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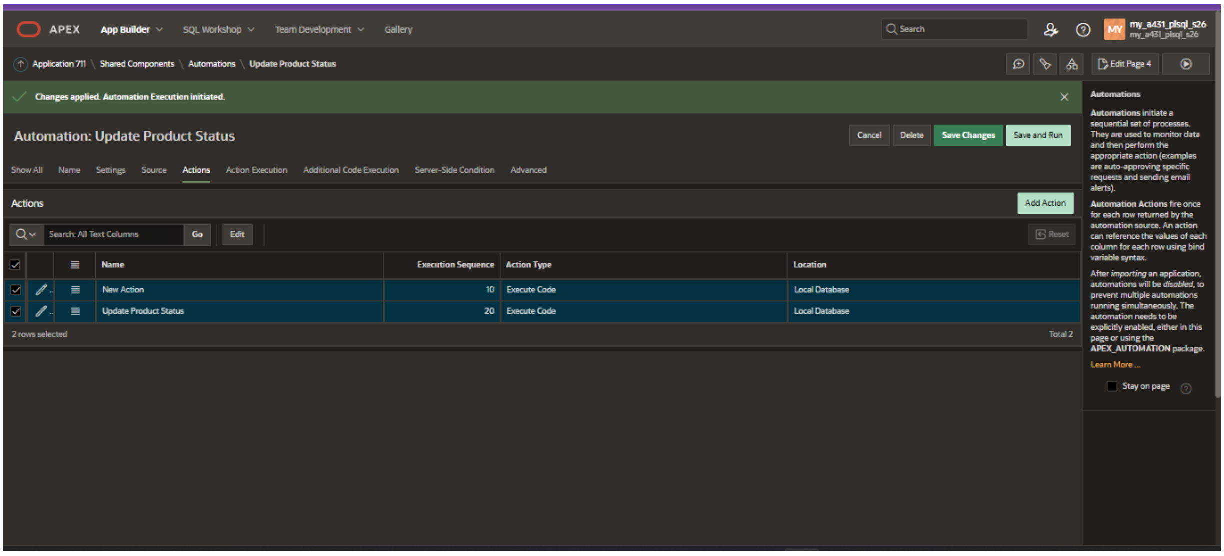
Task 2 : Create Automation

i. Automation Configuration

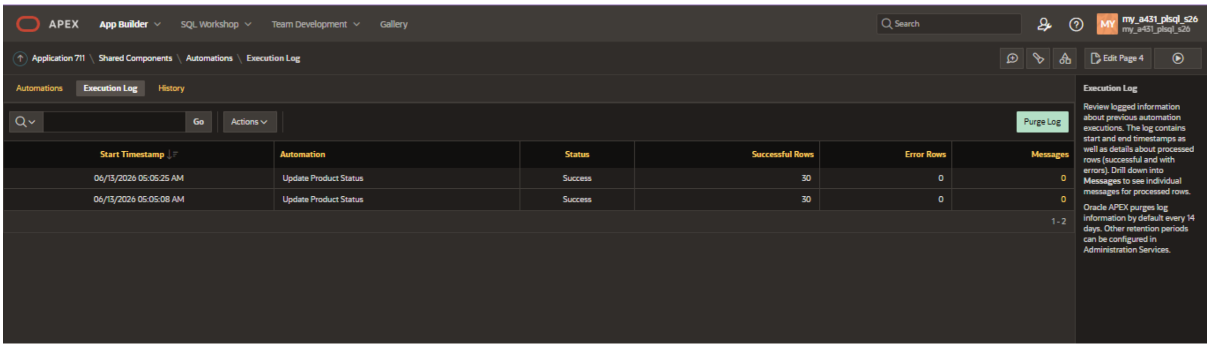
Step : app builder – share component – automations – create automation – create action



An automation was configured on the PRODUCT_SALES table to automatically update the STATUS field based on the total sales amount.

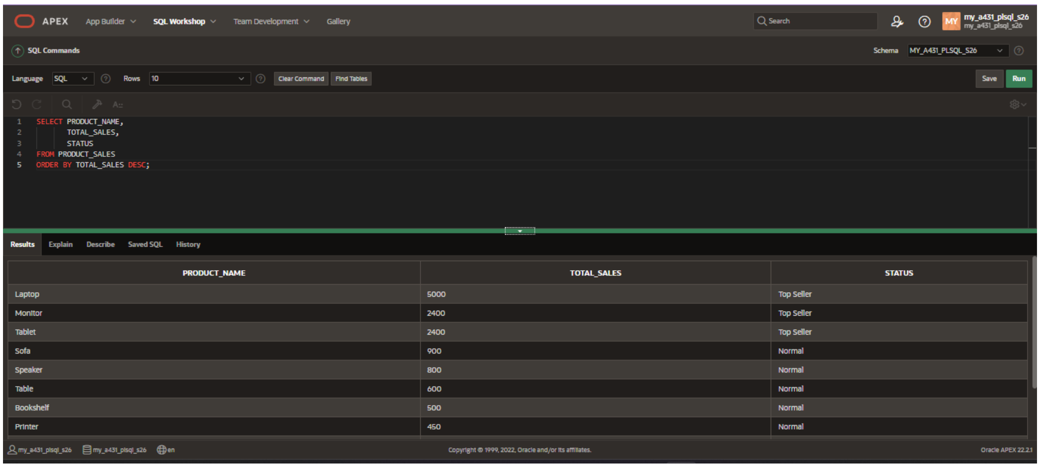
ii. Automation Execution Log

Step : share component – automation – execution log



The Automation Execution Log shows that the automation was executed successfully and processed the sales records.

Check data



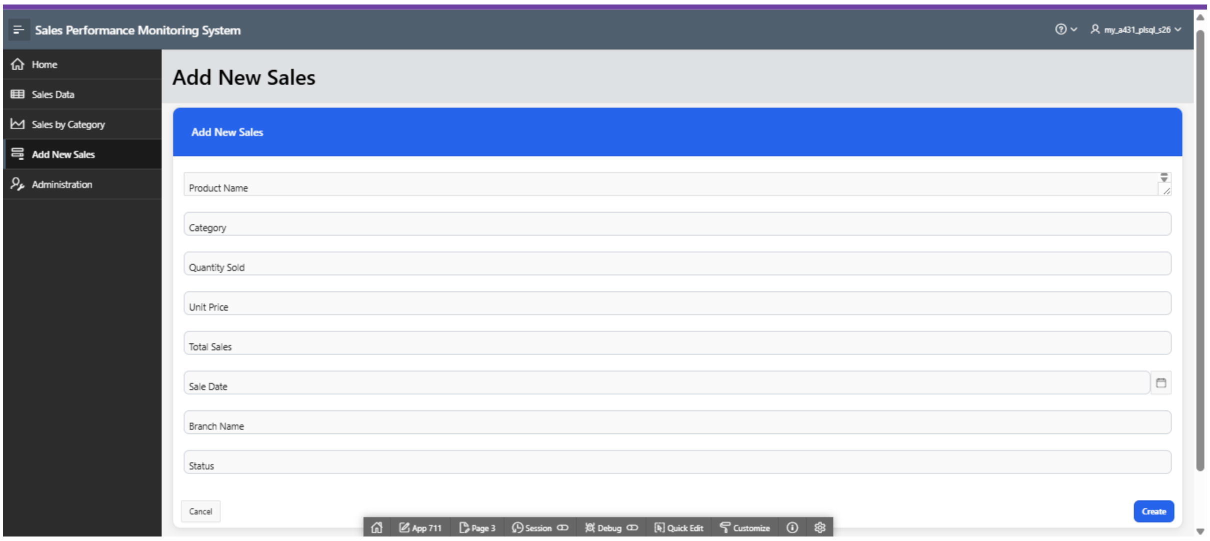
The updated sales records were verified to ensure that the STATUS field was correctly updated by the automation.

Summary Task 2 :

In this task, an automation was configured to automatically update the STATUS field in the PRODUCT_SALES table. The automation was executed successfully, and the execution log was reviewed to verify the results.

Task 3 : Create a Send Email Process

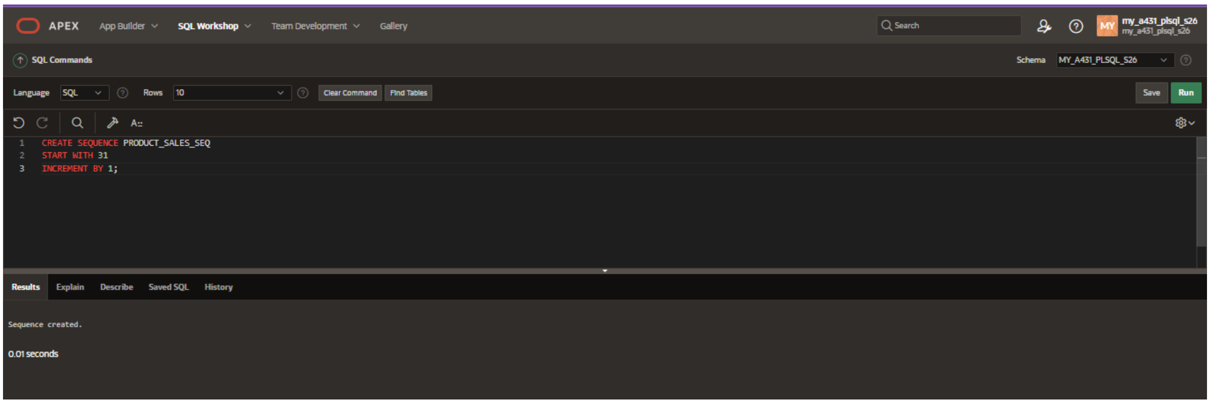
a. Create Page Form



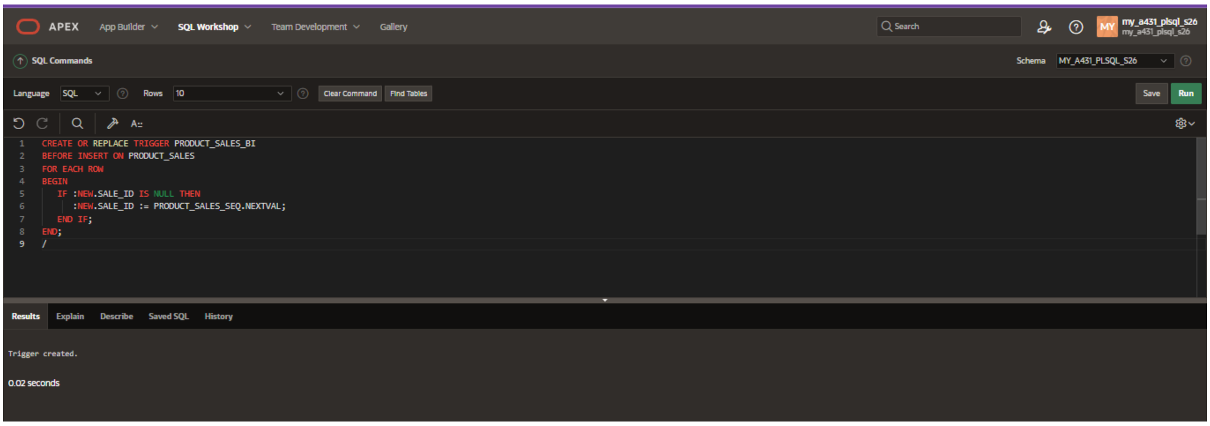
A form page named Add New Sales was created to allow users to submit new sales information into the PRODUCT_SALES table.

i. Create auto sequenced ID

A database sequence was created to automatically generate unique SALE_ID values for each new sales record.

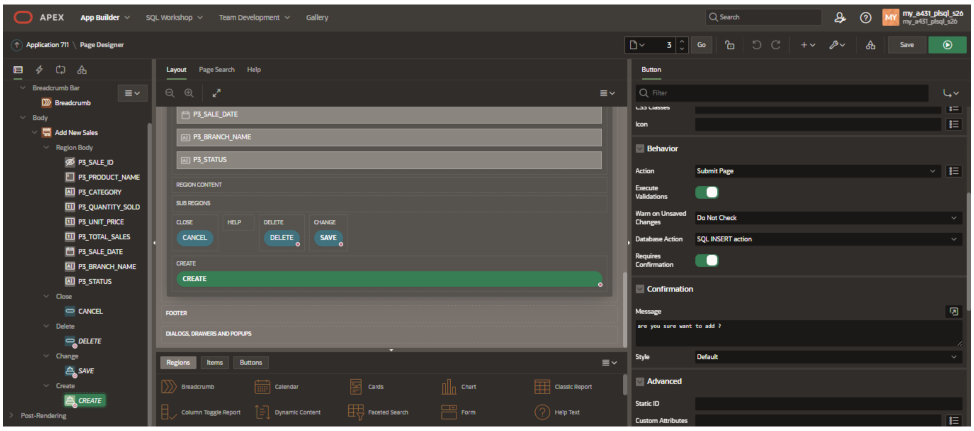


ii. Create trigger

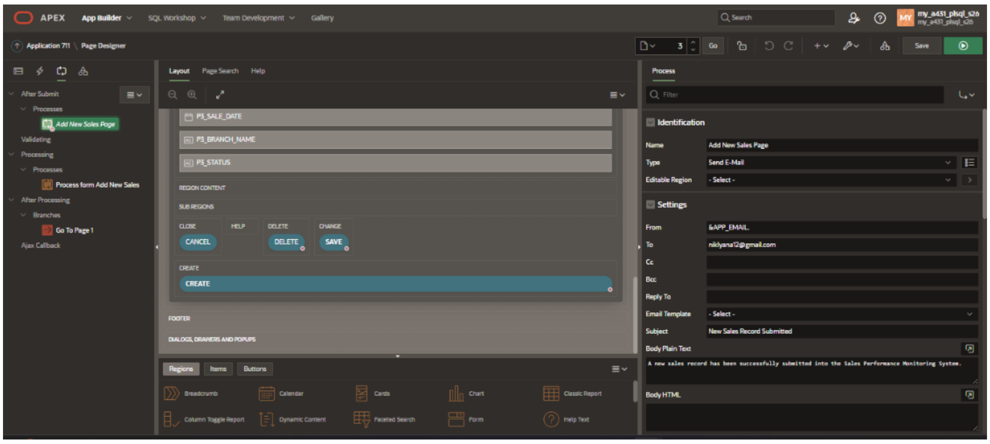


A database trigger was created to assign the next sequence value to SALE_ID before a new record is inserted.

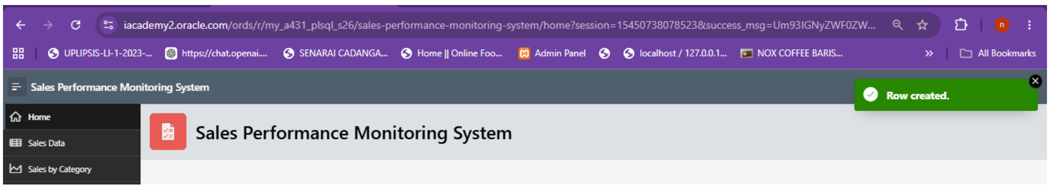
iii. Setting Page Designer + process



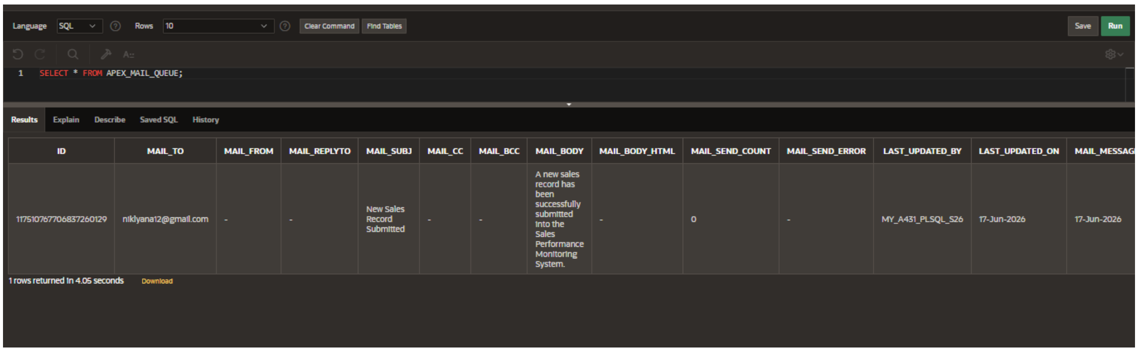
- **Add process (type = send email) setting**



iv. Row Created

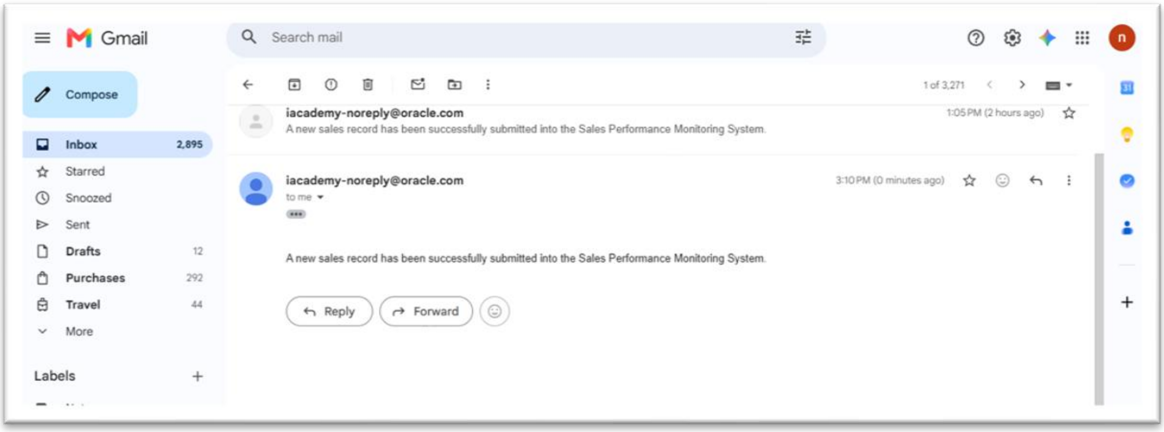


v. Configuration E-mail (queue Email)



The email queue was checked " select * from APEX_EMAIL_QUEUE;" to verify that the notification email was successfully generated and queued for delivery.

vi. Received E-mail



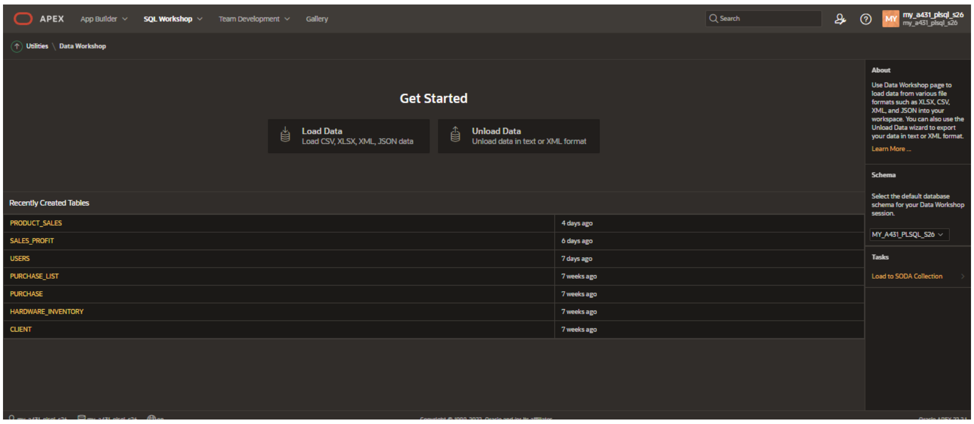
The notification email was successfully received to niklyana12@gmail.com , confirming that the Send Email process works correctly.

Summary task 3:

<u>In this task, a sales submission form was created in Oracle APEX. A Send Email process was configured to send notification emails automatically whenever a new sales record was submitted.</u>

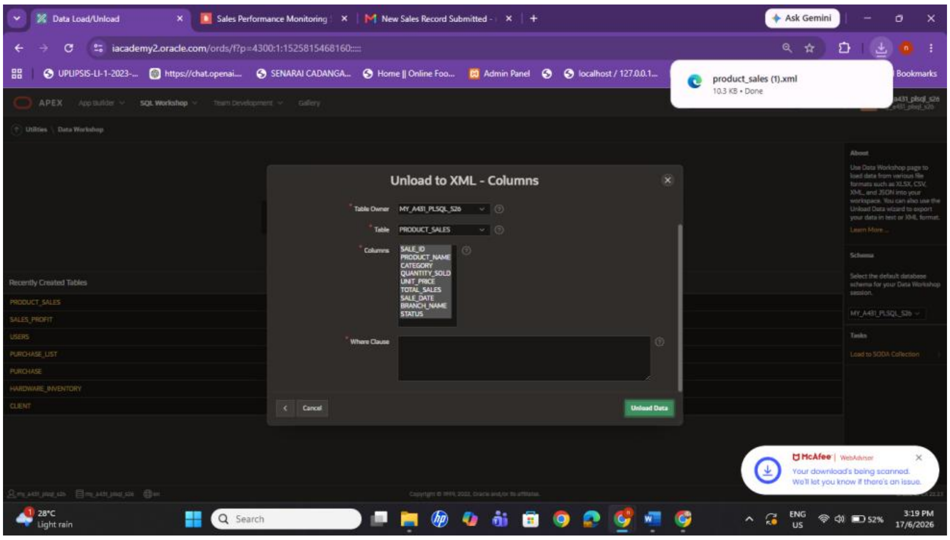
Task 4 : Migrate Database Objects and Data

PART A - Export PRODUCT_SALES Table

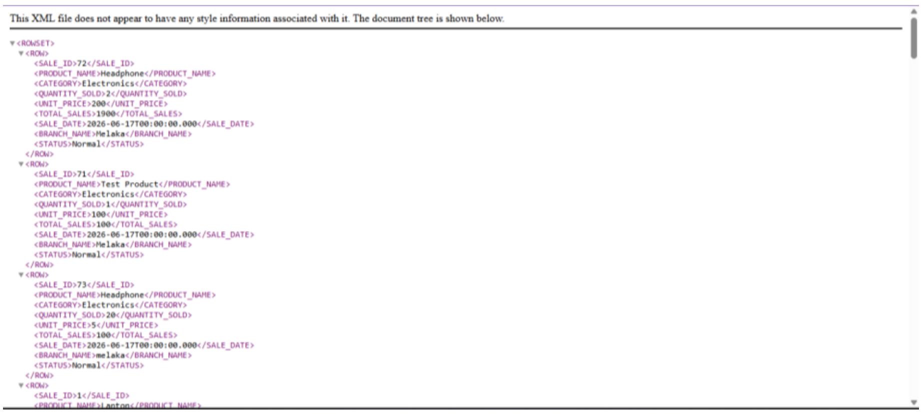


a. Unload Data

First the table data was exported using the Unload Data feature and saved in XML format.



result (xml.)



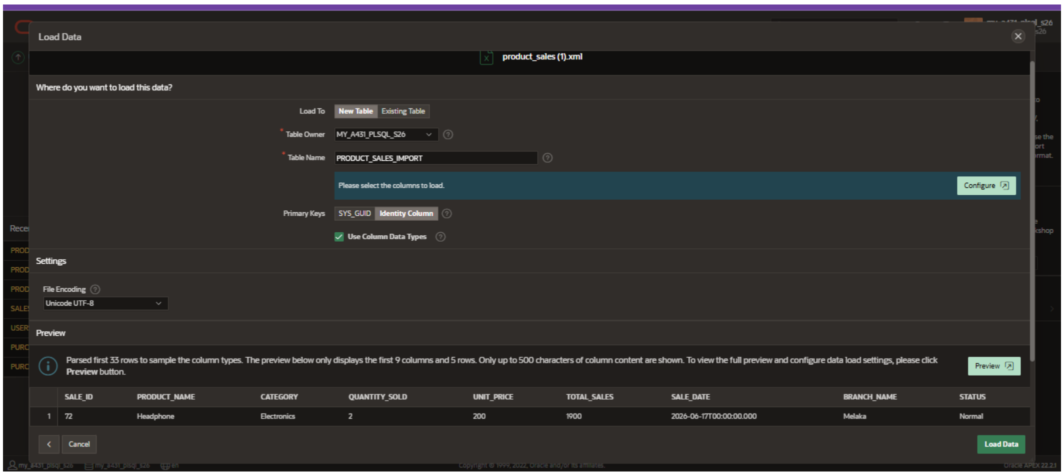
This XML file does not appear to have any style information associated with it. The document tree is shown below.

```
▼<ROWSET>
  ▼<ROW>
     <SALE_ID>72</SALE_ID>
     <PRODUCT_NAME>Headphone</PRODUCT_NAME>
     <CATEGORY>Electronics</CATEGORY>
     <QUANTITY_SOLD>2</QUANTITY_SOLD>
     <UNIT_PRICE>200</UNIT_PRICE>
     <TOTAL_SALES>1900</TOTAL_SALES>
     <SALE_DATE>2026-06-17T00:00:00.000</SALE_DATE>
     <BRANCH_NAME>Melaka</BRANCH_NAME>
     <STATUS>Normal</STATUS>
   </ROW>
  ▼<ROW>
     <SALE_ID>71</SALE_ID>
     <PRODUCT_NAME>Test Product</PRODUCT_NAME>
     <CATEGORY>Electronics</CATEGORY>
     <QUANTITY_SOLD>1</QUANTITY_SOLD>
     <UNIT_PRICE>100</UNIT_PRICE>
     <TOTAL_SALES>100</TOTAL_SALES>
     <SALE_DATE>2026-06-17T00:00:00.000</SALE_DATE>
     <BRANCH_NAME>Melaka</BRANCH_NAME>
     <STATUS>Normal</STATUS>
   </ROW>
  ▼<ROW>
     <SALE_ID>73</SALE_ID>
     <PRODUCT_NAME>Headphone</PRODUCT_NAME>
     <CATEGORY>Electronics</CATEGORY>
     <QUANTITY_SOLD>20</QUANTITY_SOLD>
     <UNIT_PRICE>5</UNIT_PRICE>
     <TOTAL_SALES>100</TOTAL_SALES>
     <SALE_DATE>2026-06-17T00:00:00.000</SALE_DATE>
     <BRANCH_NAME>melaka</BRANCH_NAME>
     <STATUS>Normal</STATUS>
   </ROW>
  ▼<ROW>
     <SALE_ID>1</SALE_ID>
     <PRODUCT_NAME>Laptop</PRODUCT_NAME>
```

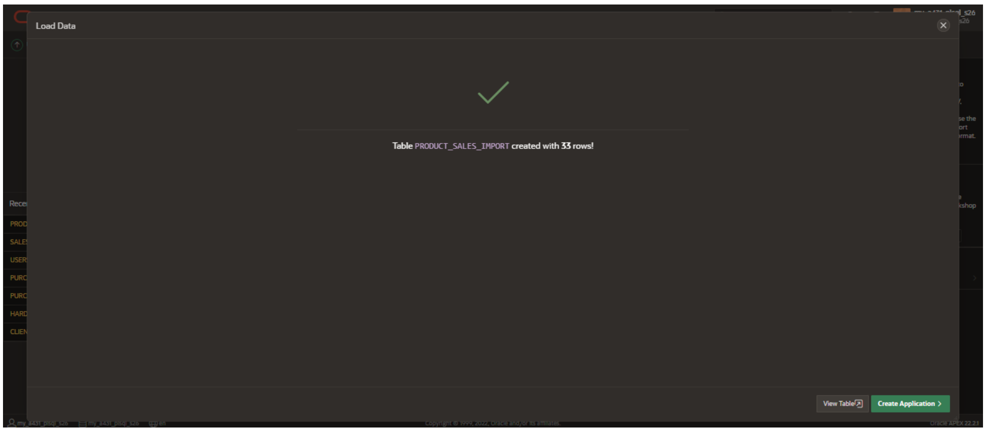
b. Load data (.CSV)

Second , using load data the exported data was imported from the export (PRODUCT_SALES) XML format to insert the new data or schema like below :

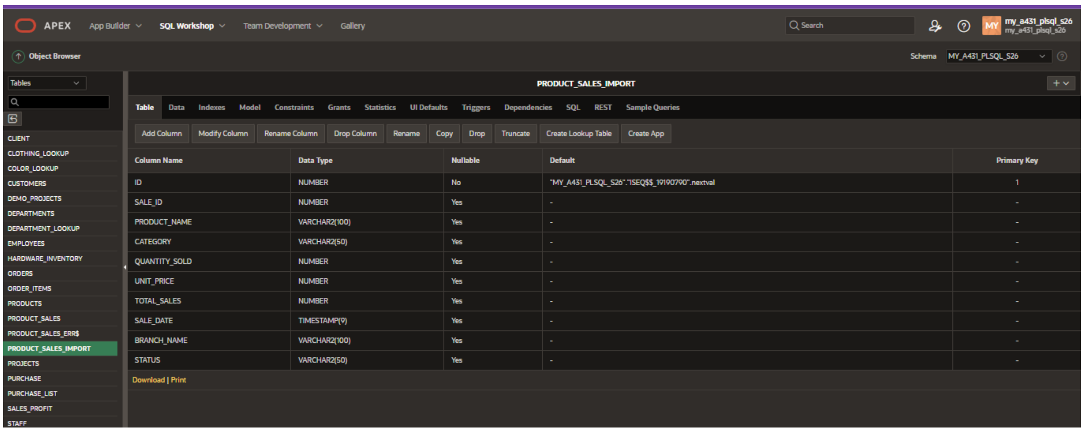
- **Rename new table**



- **Result after load data :**



c. Verify Imported Records



The imported records were verified to ensure that the data migration process was completed successfully.

Summary task 4 :

In this task, the PRODUCT_SALES table and its data were exported and imported into another environment. The migration process was successfully completed, and the imported records were verified.

Conclusion :

The Sales Performance Monitoring System was successfully developed using Oracle APEX. The system supports sales reporting, data visualisation, automation, email notification, and database migration features to improve sales monitoring and management efficiency.